Exercise 3 solution

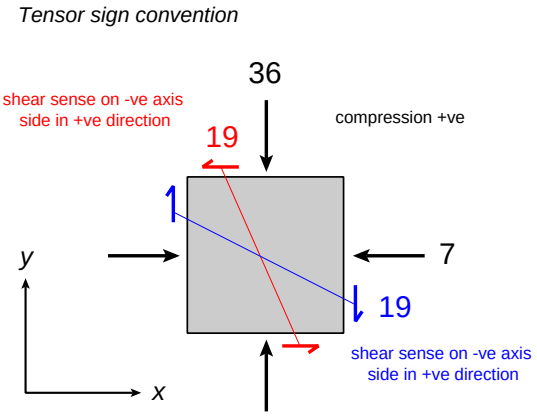
A state of stress (units MPa) is given in a Cartesian coordinate frame (x, y) by

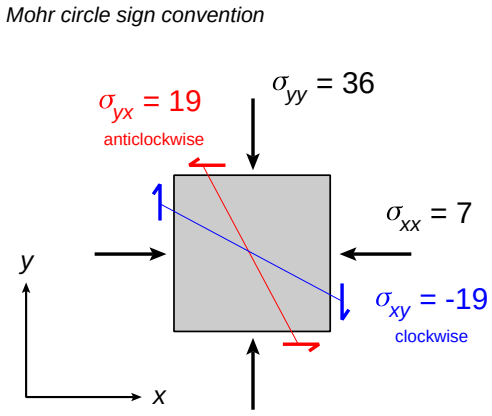
$$\begin{bmatrix} 7 & 19 \\ 19 & 36 \end{bmatrix}$$

Part (a)

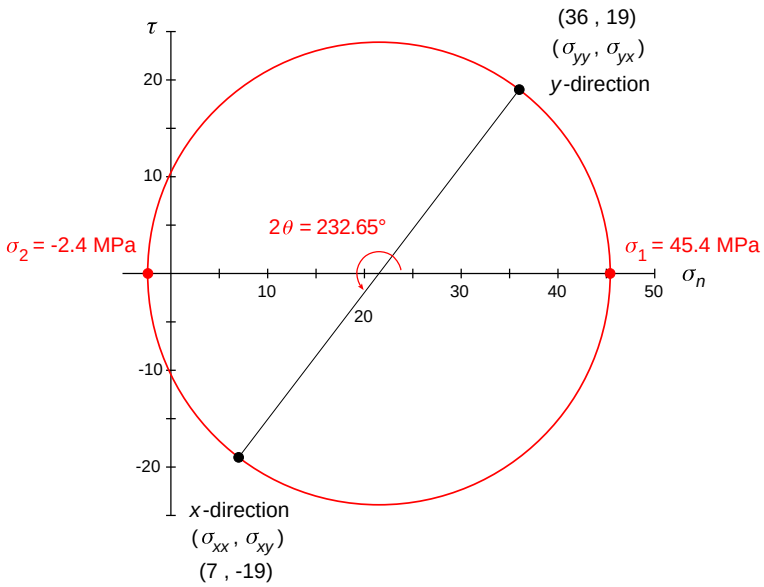
Using a Mohr diagram, solve for the principal stresses, σ_1 and σ_2, and the orientation, θ, between σ_1 and the x-direction.

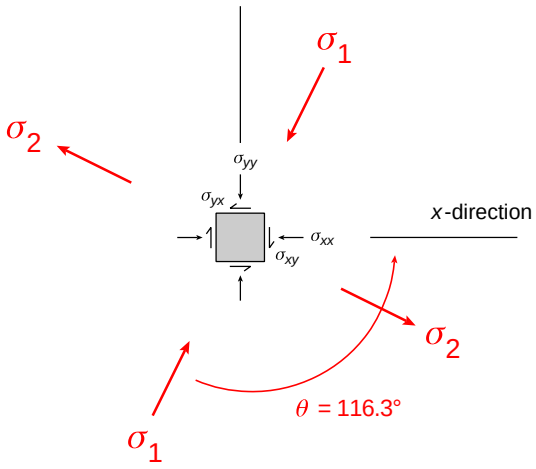
The state of stress is presented in tensor sign convention, and so first task is to determine the sign of the shear stresses using the Mohr circle convention. We draw a square element and orient the stresses in an (x, y) coordinate frame, noting that since the shear stresses are positive, for any given shear couple, on the side that cuts the negative axis the shear stress should be going in a positive direction. We then determine the sign of the shear stresses in Mohr circle convention (for given shear couple, clockwise is negative and anticlockwise is positive).





We can now draw the Mohr circle and read off the answers.





Part (b)

Check your answers to (a) using the formulae,

$$\sigma_1 , \sigma_2 = \frac{(\sigma_{xx} + \sigma_{yy})}{2} \pm \left[\frac{(\sigma_{xx} - \sigma_{yy})^2}{4} + \sigma_{xy}^2 \right]^{½} \quad ; \quad \tan 2\theta = \frac{2\sigma_{xy}}{(\sigma_{xx} - \sigma_{yy})}$$

With

$$\sigma_{xx} = 7 \text{ MPa} \quad ; \quad \sigma_{yy} = 36 \text{ MPa} \quad ; \quad \sigma_{xy} = 19 \text{ MPa}$$

then

$$\sigma_1 = (43/2) + [(-29^2/4) + 19^2]^{½} = 45.4 \text{ MPa}$$

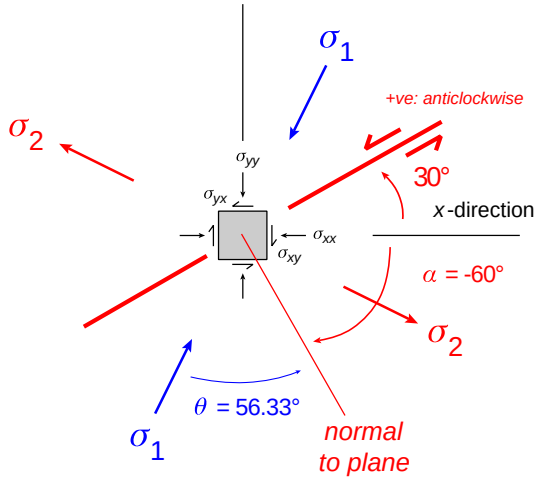
$$\sigma_2 = (43/2) - [(-29^2/4) + 19^2]^{½} = -2.4 \text{ MPa}$$

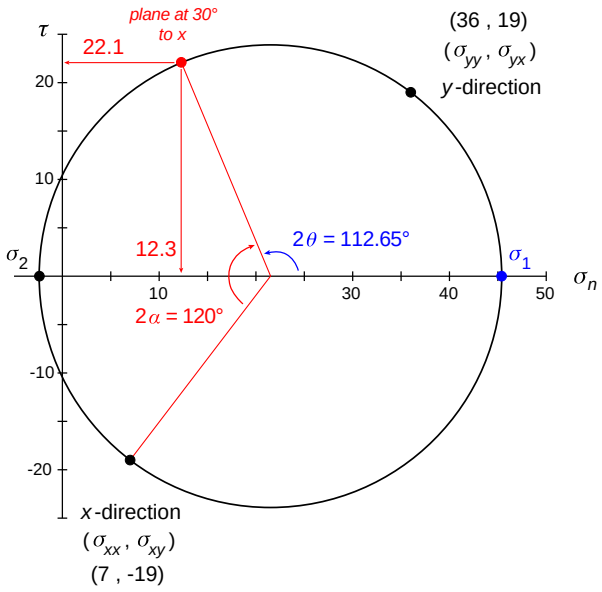
$$\tan 2\theta = -38/(-29) \rightarrow \theta = 26.3° \text{ or } 116.3°$$

With the value of θ, note that in the range 0-180° there are two values of θ that satisfy the $\tan 2\theta$ equation (one finds the angle to σ_1 and the other to σ_2). These differ by 90° (26.3 + 90 = 116.3°). The rules for selecting the correct one are listed in the notes supporting the stress lecture (see also Jaeger *et al.*, 2007, *Fundamentals of Rock Mechanics* (4th edition), Blackwell, p.21). In this case $\sigma_{xx} < \sigma_{yy}$ and $\sigma_{xy} < 0$ (Mohr circle sign convention) and so the angle between the *x*-direction and σ_1 lies between 90° and 135°. Hence 116.3° is the correct solution.

Part (c)

Using your Mohr diagram, determine the value of normal stress (σ_n) and shear stress (τ) on a plane oriented at 30° anticlockwise to the x direction.





Part (d)

Check your answers to (c) using the formulae,

$$\sigma_n = \frac{(\sigma_1 + \sigma_2)}{2} + \frac{(\sigma_1 - \sigma_2)}{2} \cos 2\theta \quad ; \quad \tau = \frac{(\sigma_1 - \sigma_2)}{2} \sin 2\theta$$

With

$$\sigma_1 = 45.4 \text{ MPa} \quad ; \quad \sigma_2 = -2.4 \text{ MPa} \quad ; \quad \theta = 56.33°$$

then

$$\sigma_n = (43/2) + [(47.8/2) \cos (112.65)] = 12.3 \text{ MPa}$$

$$\tau = (47.8/2) \sin (112.65) = 22.1 \text{ MPa}$$

Part (e)

Solve for σ_1, σ_2, and θ by finding the eigenvalues of the stress matrix given above and the eigenvector associated with σ_1.

We seek to convert our original stress matrix into a diagonal equivalent:

$$\begin{bmatrix} 7 & 19 \\ 19 & 36 \end{bmatrix} \longrightarrow \begin{bmatrix} \sigma_1 & 0 \\ 0 & \sigma_2 \end{bmatrix}$$

In doing this we seek the eigenvalues and eigenvectors of the matrix so that

$$\boldsymbol{A}\boldsymbol{v} = \lambda\boldsymbol{v}$$

where $\boldsymbol{A}$ is a square matrix (here our stress matrix), and $\boldsymbol{v}$ is the eigenvector associated with eigenvalue λ. We can rewrite this equation as

$$(\boldsymbol{A} - \lambda\boldsymbol{I})\boldsymbol{v} = 0$$

where $\boldsymbol{I}$ is the unit matrix. For non-zero eigenvector, this requires that the determinant, $|\boldsymbol{A} - \lambda\boldsymbol{I}| = 0$.

• First we find the matrix, $\boldsymbol{A} - \lambda\boldsymbol{I}$

$$\boldsymbol{A} - \lambda\boldsymbol{I} = \begin{bmatrix} 7 & 19 \\ 19 & 36 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} = \begin{bmatrix} 7 - \lambda & 19 \\ 19 & 36 - \lambda \end{bmatrix}$$

• Then we find the determinant of this matrix and set it equal to zero

Recall $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$

$$\left|\boldsymbol{A} - \lambda\boldsymbol{I}\right| = \begin{vmatrix} 7 - \lambda & 19 \\ 19 & 36 - \lambda \end{vmatrix} = (7 - \lambda)(36 - \lambda) - 19^2 = 0$$

$$= \lambda^2 - 43\lambda - 109 = 0$$

• Using the quadratic formula we then solve for the two values of λ

Recall $x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

$$\lambda^2 - 43\lambda - 109 = 0 \longrightarrow \lambda = \frac{43 \pm \sqrt{43^2 + 436}}{2}$$

$$\lambda = 45.4 \text{ or } -2.4$$

• We now seek the eigenvector corresponding to the eigenvalue of 45.4:

$$0 = (\boldsymbol{A} - \lambda \boldsymbol{I})\boldsymbol{v} = \begin{bmatrix} 7-\lambda & 19 \\ 19 & 36-\lambda \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -38.4 & 19 \\ 19 & -9.4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

$$\begin{aligned} -38.4\,x + 19\,y &= 0 \\ 19\,x - 9.4\,y &= 0 \end{aligned} \longrightarrow \begin{aligned} -2.02\,x + y &= 0 \\ y &= 2.02\,x \end{aligned}$$

$$\text{eigenvector} \quad \begin{bmatrix} 1 \\ 2.02 \end{bmatrix}$$

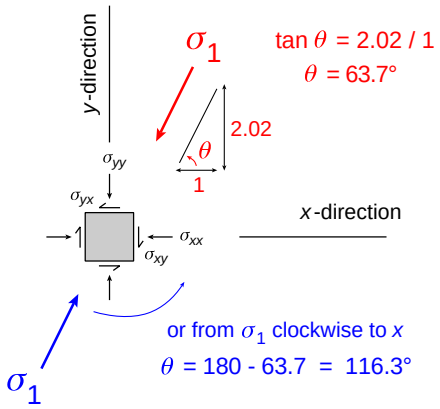
check:

$$\boldsymbol{A}\ \boldsymbol{v} = \lambda\ \boldsymbol{v}$$

$$\begin{bmatrix} 7 & 19 \\ 19 & 36 \end{bmatrix} \begin{bmatrix} 1 \\ 2.02 \end{bmatrix} = 45.4 \begin{bmatrix} 1 \\ 2.02 \end{bmatrix}$$

$$\begin{bmatrix} 45.4 \\ 91.7 \end{bmatrix} = \begin{bmatrix} 45.4 \\ 91.7 \end{bmatrix}$$

• Now to convert this eigenvector into the angle between σ_1 and the x-direction:



• The eigenvector corresponding to the eigenvalue of -2.4 is at 90° to this.